JIAYAN (STEPHEN) XIE

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EDUCATION

Duke Kunshan University **Kunshan, China**

B.S. in Molecular Bioscience / Cell and Molecular Biology *Aug 2023 – May 2027 (Expected)*

Cumulative GPA: 3.45/4.0 | Junior-Year GPA: 3.8/4.0

Duke University **Durham, NC, USA**

B.S. in Interdisciplinary Studies (Molecular Bioscience / Cell and Molecular Biology), Dual Degree Program

Honors & Awards: Dean's List, Duke University (Spring 2026) — Semester GPA 3.925; iGEM Silver Medal (Oct 2024), Team Leader; DKU Undergraduate Scholarship, Duke Kunshan University (Aug 2023) — 170,000 RMB

RESEARCH EXPERIENCE

TUBGCP2 Functional Study in Prostate Cancer Cells

Huang Lab, DKU | Supervisor: Prof. Linfeng Huang | Signature Work Independent Project *Mar 2026 – Present*

Overview: This project uses siRNA knockdown in prostate cancer cells to assess whether TUBGCP2 is a cancer dependency gene, with BIRC5 as a benchmarking control to distinguish biological effects from technical artifacts.

- Designed three independent siRNAs targeting *TUBGCP2* and BIRC5 in PC-3 and DU145 cells; systematically evaluated knockdown efficiency via qPCR and Western blot
- Established a multi-layered control framework: FAM-siRNA (delivery control), GAPDH-siRNA (knockdown workflow control), KIF11-siRNA (phenotypic positive control), and NC-siRNA (negative baseline) for systematic troubleshooting of the experimental system
- Planned functional assays upon confirmation of knockdown, including CCK-8 cell viability, cell proliferation counting, and morphological imaging
- Current status: Optimizing qPCR conditions and transfection efficiency; KIF11 positive control shows reduced cell count and morphological abnormalities, while *TUBGCP2* knockdown groups have not yet exhibited obvious phenotypes

Functional Genomics in Insect Cell Models to Study Cellular Pathways

Huang Lab, DKU | Supervisor: Prof. Linfeng Huang | Summer Research Scholars Program *Jun 2026 – Oct 2026*

Overview: This project extends pro-siRNA technology to insect cell models (S2, Sf9/Sf21, C6/36) and combines it with high-content imaging to establish a phenotypic screening platform for identifying genes regulating cell viability, RNA delivery, and gene silencing.

- Contributed to preparing over 30 96-well pro-siRNA library plates; performed high-content imaging-based quantification of dead/live cell ratios at 72 hours post-transfection, identifying multiple candidate wells with altered cell viability
- Developed an eGFP fluorescent reporter system; compared PEI and Lipofectamine 3000 transfection reagents; achieved ~30% transfection efficiency with Lipofectamine 3000; detected sequence-specific gene silencing via co-transfection of eGFP mRNA with eGFP-targeting siRNA
- Established a complete workflow for quantifying cell viability changes, RNA delivery efficiency, and gene silencing, successfully extending pro-siRNA technology from mammalian cancer cells to insect cell models

Construction and Verification of Long Hairpin RNA Plasmids for Recombinant siRNA Production in *E. coli*

Huang Lab, DKU | Supervisor: Prof. Linfeng Huang | Summer Research Scholars Program *Apr 2025 – Nov 2025*

Overview: This project uses E. coli to produce recombinant siRNA via long hairpin RNA plasmids, providing a more efficient and cost-effective platform for high-throughput RNAi screening.

- Constructed long hairpin RNA plasmids for 17 cancer-related genes by selecting 200–689 bp regions downstream of ATG start codons and performing two-step sequential cloning (SacI/XhoI + NotI/SalI)
- Completed full molecular cloning workflow: high-fidelity PCR, restriction digestion, ligation, transformation, colony screening, plasmid extraction, and Sanger sequencing
- Optimized digestion conditions (extended incubation from 1.5 to 2 hours, doubled enzyme and DNA input), improving cloning efficiency and reproducibility

- Detected ~21 nt siRNA products via RNA gel electrophoresis with consistent 260 nm UV absorption peaks; completed validation for all 17 constructs

Tampon X: Engineered Lactic Acid Bacteria-Based System for Prevention of Tampon-Associated Toxic Shock Syndrome *Dec 2023 – Oct 2024*

DKU BSL-2 Laboratory, DKU iGEM Team | Nisin Wet Lab Group Leader | Supervisor: Prof. Linfeng Huang

Overview: This project engineers Lactococcus lactis (nisin enhancement) and Lactobacillus johnsonii (H_2O_2 enhancement) to inhibit Staphylococcus aureus and prevent Toxic Shock Syndrome, exploring microbiome-based preventive strategies.

- Led the Nisin group and directed the promoter screening module: tested 14 lactic acid bacterial promoters; identified PnisA and PTSIIC-celA as strong expression promoters via ELISA and Western blot
- Constructed pMG36e-AcGFP reporter vectors; validated that NisRK overexpression enhances PnisA-driven expression by increasing GFP signal
- Contributed to CRISPR/Cas9 genome editing strategy design for replacing the native PnisR promoter with a strong promoter to boost nisin production
- Team achievements: Completed six DBTL iteration cycles; constructed and verified an ~12 kb pMG-Cas vector via PCR and single-enzyme digestion; H_2O_2 route achieved a peak of 41.2 μM under anaerobic conditions with LJ_0548/LJ_0549 overexpression; awarded iGEM Silver Medal 2024

SELECTED ACADEMIC PROJECTS

Micro/Nanoplastic Neurotoxicity Mechanism Study *Feb 2025 – Apr 2025*

- Systematically reviewed the molecular pathways by which micro/nanoplastics translocate from the gut to the bloodstream, cross the blood-brain barrier, and induce mitochondrial dysfunction
- Constructed a causal framework linking environmental exposure (ROS overproduction, ATP depletion) to neurodegenerative phenotypes
- Compared size-dependent effects (micro-scale vs. nano-scale) on biological barrier penetration and cytotoxicity, identifying nanoplastics as posing a more direct threat at the organelle level due to their ability to enter mitochondria

AI-Powered Multi-Modal Clinical Decision Support System *Oct 2025 – Nov 2025*

- Designed a conceptual framework for an AI-based diagnostic tool integrating genetic, imaging, and clinical data to address pharmacogenetic challenges, such as CYP2D6-guided analgesia and DPYD-guided fluoropyrimidine dosing
- Conducted pharmacogenetics background research across osteoarthritis, cardiovascular, and oncology contexts; evaluated cross-disease applicability
- Proposed an iterative model training framework incorporating standardized clinical workflows and AI optimization through successive prediction–adjustment cycles; provided preliminary assessment of commercialization potential

EXTRACURRICULAR ACTIVITIES

BIOL 304 Molecular Genetic Analysis *Oct 2026 – Present*

Teaching Assistant / Course Instructor: Prof. Linfeng Huang

- Facilitate classroom discussions, review sessions, and provide support for undergraduate students in molecular genetics
- Assist the course instructor in grading undergraduate assignments, quizzes, and final examinations

Varsity Ultimate Frisbee Team

Spirit Captain *Jan 2025 – Present*

- Led pre-tournament rules briefings and coordinated in-game communication among teammates and opponents to clarify calls and resolve disputes
- Facilitated post-match debriefs and reinforced fair play and sportsmanship, earning multiple Best Spirit Awards

SKILLS

Laboratory Skills: *Mammalian and insect cell culture, siRNA transfection and knockdown, RNA immunoprecipitation (RIP), qPCR primer design and validation, gradient PCR optimization, agarose and RNA gel electrophoresis, RT-qPCR, Western blot (SDS-PAGE, membrane transfer, and immunodetection), ELISA, molecular cloning (PCR, restriction digestion, ligation, transformation, plasmid extraction), Sanger sequencing analysis, high-content imaging, flow cytometry, CCK-8 cell viability assay*

Software: *RStudio, GraphPad Prism, Primer3*

Languages: *Chinese (Native), English (Fluent)*